Shouhao Chang

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Education

Hunan University Sep 2024 – Jun 2027 (expected)
M.S. in Computer Science *Changsha, China*

- Admitted by recommendation and exempt from the entrance exam.

Hunan University Sep 2020 – Jun 2024
B.S. in Computer Science *Changsha, China*

- National College Entrance Examination (Henan): ranked 7,000th among 1.15 million candidates.

Research Experience

R^2E: A Role-driven Reward Evolutionary Framework for Automated Reward Function Design
Sole first author Accepted at NeurIPS 2026

- Reward design in reinforcement learning still relies on costly manual trial and error. LLM-based methods explore the reward space poorly and make training unstable, because candidates are sampled without distinction and past feedback is underused.
- R^2E splits the search across three LLM roles: an Explorer for novel rewards, a Guardian for conservative edits that stabilize training, and an Evolver that recombines strong rewards by crossover and mutation. A scheduler shifts from exploration to exploitation, and a Global Elite Pool keeps the best rewards so the roles can build on them.
- Experiments on multiple robotic tasks show that the generated reward functions are effective.

Parameter-Free Historical Visual Memory Enhancement for VLA Models Target for CVPR 2027

- Current PI models act only on the present frame and have no visual memory across time. In tasks such as placing a bottle into a paper bag, several similar handles appear at once, and the models cannot use earlier views to choose the right one. The goal is to give vision-language-action (VLA) models this memory.
- The memory has to be added without hurting the original tasks. Editing the SigLIP encoder with new learnable parameters broke the existing visual-token semantics and lowered real-robot performance. Inspired by PI-MEM, the method leaves the architecture unchanged: the same encoder embeds the current frame and past frames, spatiotemporal attention mixes their tokens, and only the current-frame output is passed to the LLM, so the original VLA interface stays intact.
- Early real-robot trials show a gain on these tasks, and simulation experiments are underway. For any VLA model, historical visual memory can be added by post-training alone, without new pretraining.

Internship

ACE Robotics May 2026 – Present
Research Intern *Shenzhen, China*

- Post-trained $\pi_{0.5}$ and deployed it on real robots. Collected teleoperation data, checked its quality, and converted the format, then ran SFT and offline training, followed by online inference and iteration on the robot. When the robot failed, traced the cause back to the data and corrected it.
- At the World Artificial Intelligence Conference (WAIC), trained the model and fixed bugs in real time, and received a WAIC special incentive.

Skills

Research	Reinforcement learning, deep learning, and robotics.
Engineering	Python, C++, PyTorch, JAX, Git workflow, and real-robot deployment.
Personal	Responsible, clear in communication, and reliable in delivery.